Chirality and Charge Quantization from the Two Arenas

The Weak Factor is Chiral Because It Is the Left Factor; Electric Charge Comes in Thirds Because the Clock Unit Grades the Octonions

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Abstract

Paper LVI derived the gauge inventory available to the rendered VFD substrate — $SU(3) \times SU(2)$ with a distinguished clock circle — and listed, among its honest opens, chirality and the charge/representation wiring. This paper closes both at structure grade, using no ingredients beyond those the programme has already forced.

Chirality. The arena's rotation group is $\text{Spin}(4) = SU(2)_{\text{left}} \times SU(2)_{\text{right}}$, and these factors are literally left and right unit-quaternion multiplication — the same two actions whose roles the rendered frame split already fixed: right = the scene's spatial rotations, left = internal. The new observation is representation-theoretic: those are exactly the two *chirality* factors. Fixing the complex structure $J = R_i$, the quaternions become the 2-complex-dimensional Weyl module; every left multiplication is J-linear and acts as $SU(2)$ (the standard $2\mathbf{I} \hookrightarrow SU(2)$, verified for all 120 substrate units), while right multiplication is J-linear only on the four units of the i-line: *the spatial factor cannot act on the chiral module at all.* An internal gauge group inherited from the left factor therefore couples to one Weyl sector only — the handedness of the weak interaction is the same fact as the frame split. Stated with its caveats: this is the Euclidean $\text{Spin}(4)$ statement, surviving the standard continuation to $SL(2, \mathbb{C})$ as the $(\frac{1}{2}, 0)/(0, \frac{1}{2})$ split, and it derives why a left-factor group is chiral, not the full electroweak fermion sector.

Charge quantization. On the complexified octonions, with the same clock unit e_1 and complex structure $J = L_{e_1}$ that break $G_2 \to SU(3)$ in Paper LVI: the transverse space splits into exactly three J-complex lines; the ladder operators built on them satisfy the fermionic algebra $\{a_i, a_j^\dagger\} = \delta_{ij}$, $\{a_i, a_j\} = 0$ exactly; the number operator has spectrum $\{0, 1, 2, 3\}$ with multiplicities $\{1, 3, 3, 1\}$; and the

$\mathfrak{su}(3)$ stabiliser commutes with it, and its restricted image on each 3-dimensional eigenspace is the full 8-dimensional algebra — faithful, hence (by the simplicity established in Paper LVI and the absence of nontrivial $\mathfrak{su}(3)$ representations of dimension < 3) exactly the triplet — while it vanishes on the 1-dimensional ones. With $Q = N/3$: a charge-*magnitude* lattice quantized in thirds, in the pattern of one neutral singlet, one charge-$\frac{1}{3}$ colour triplet, one charge-$\frac{2}{3}$ colour triplet, one charge-1 singlet — the magnitudes of $(\nu, \bar{d}, u, e^+)$. A genuine null: a non-J-compatible pairing destroys the integer spectrum. The construction is classical (Günaydin–Gürsey 1973; Furey 2016); the programme's addition is necessity at the input layer — this algebra and this distinguished unit are forced by the arena theorems, not chosen. A corollary fixes the bookkeeping half of hypercharge ($Y = 2(Q - T_3)$ once doublet assignments are made); the independent gauging of $U(1)_Y$, the Weinberg angle, the generation count (named here as the D_4-triality conjecture), and all masses remain open and are listed.

1 Position

Paper LVI [Smart, 2026] ended with five honest opens for the Standard-Model bridge: the embedding pattern (independent hypercharge / Weinberg angle), chirality, representation content, couplings and masses, and non-abelian uniqueness provenance. This paper closes the second and the structural core of the third. The method is the programme's usual one: no new objects — both results use only the two arenas, the frame split, and the clock unit, each already forced and verified upstream — plus machine verification with falsifiable nulls for every claim (suite `verify_sm_structure.py`; items cited inline).

2 Chirality: the left factor is the chiral factor

2.1 The observation

Three facts, the first two already in the record:

(i) On the quaternionic arena, the 4D rotation group is $\mathrm{Spin}(4) = SU(2)_{\mathrm{left}} \times SU(2)_{\mathrm{right}}$, acting by $x \mapsto q\,x\,\bar{r}$ — left and right unit-quaternion multiplication.

(ii) The rendered frame split (Papers LIII/LVI, verification items T14a–b): the *right* action rotates the canonical observer frame — it is the scene's spatial rotation group, geometrically spent — while the *left* action transports frame components trivially and is the arena's internal symmetry.

(iii) (New here.) The two factors of $\mathrm{Spin}(4)$ are its two *chirality* factors: its Weyl (half-spinor) representations are $(\mathbf{2}, \mathbf{1})$ and $(\mathbf{1}, \mathbf{2})$ — each factor acts on one handedness and is blind to the other.

Combining (ii) and (iii): *the internal symmetry the substrate hands to the gauge sector is a chirality factor.* Whatever it couples to, it couples one-handedly — not by a choice in the matter sector, but because the other handedness is the sector the scene's own rotations act on.

2.2 The verified module statement

Proposition 1 (The Weyl module and who can act on it)**.** *Let $J = R_i$ (right multiplication by i) on $\mathbb{H} \cong \mathbb{R}^4$. Then:*

(a) $J^2 = -1$, and every left multiplication L_q commutes with J: the pair $(\mathbb{H}, J)$ is a 2-complex-dimensional module on which the left factor acts $\mathbb{C}$-linearly (item CH1; exact, all 120 substrate units).

(b) In the complex basis $\{1, j\}$, $q \mapsto L_q$ is the standard inclusion $2\mathbf{I} \hookrightarrow SU(2)$: each matrix is unitary with determinant 1, and the assignment is a group homomorphism (item CH2).

(c) For the continuous *right factor: R_q commutes with $J = R_i$ iff $qi = iq$ iff q lies in the complex line of i — a one-line statement holding for all unit quaternions, so the C-linear part of $SU(2)_{\mathrm{right}}$ on this module is exactly the circle $U(1)_i$, and no non-abelian part of the spatial factor acts on it. The substrate witness: among the 120 icosian units exactly $\{\pm 1, \pm i\}$ commute with J (item CH3).*

(d) The left action is irreducible: it is unitary and 2-dimensional, so reducibility would force a sum of two characters and an abelian image, while the image contains non-commuting elements (item CH4).

(e) The mirror module exists and swaps the roles exactly: with the left *complex structure $J_L = L_i$, every right multiplication is C-linear and left multiplication is C-linear only on the i-line (item CH5). Each* Spin(4) *factor owns exactly one of the two Weyl modules — the precise sense of statement (iii) above.*

Proof. (a) $J^2 = R_{i^2} = R_{-1} = -\mathrm{id}$; left and right multiplications commute by associativity. (b) Writing $x = 1 \cdot z_1 + j \cdot z_2$ with right-scalar action, L_q is $\mathbb{C}$-linear by (a), and unitarity/determinant/multiplicativity are verified exhaustively. (c) $R_q J = J R_q$ iff $R_{qi} = R_{iq}$ iff $qi = iq$, i.e. $q \in \mathrm{span}(1, i)$; intersecting with the substrate units gives $\{\pm 1, \pm i\}$, as verified. (d) As argued in the statement: unitary 2-dim plus non-abelian image. (e) The mirror of (a)–(c) under left–right exchange, verified exhaustively. □ □

Corollary 2 (Chirality of the internal factor)**.** *An internal gauge group obtained from the left factor (Paper LVI's $SU(2)$, with its clock circle) acts $\mathbb{C}$-linearly on the Weyl module of Proposition 1 and trivially on the conjugate module; matter charged under it is charged one-handedly. In Lorentzian continuation the split becomes the $(\frac{1}{2}, 0)/(0, \frac{1}{2})$ decomposition of $SL(2, \mathbb{C})$ spinors, and the statement survives as: the internal factor couples to left-handed Weyl components.*

Remark 3 (Caveats, in full). The derivation is at the level of the arena's Euclidean spin group and the module structure; the Wick continuation invoked in Corollary 2 is the standard one but is imported, not re-derived. Nothing here constructs the electroweak fermion Lagrangian, fixes doublet assignments, or addresses anomaly bookkeeping (§5). What is derived is the mechanism question the honest list posed: *why would the substrate's internal $SU(2)$ be chiral at all?* Answer: because it is the left factor, and the right factor is the sky.

3 Charge: the clock unit grades the octonions in thirds

3.1 Construction

All objects are the ones Paper LVI already used: the octonions $\mathbb{O}$ built by Cayley–Dickson doubling (norm composition verified), the clock unit e_1, the complex structure $J = L_{e_1}$ with $J^2 = -1$ on the transverse space $W = \mathrm{span}(e_2, \dots, e_7)$.

Proposition 4 (Ladder algebra)**.** *W splits into exactly three J-complex lines (f_k, Jf_k), $k = 1, 2, 3$ (item Q0). The operators*

$$a_k \;=\; \tfrac{1}{2}\big(L_{f_k} - i\, L_{Jf_k}\big)$$

on $\mathbb{O}\otimes\mathbb{C}$ satisfy the canonical fermionic anticommutation relations exactly: $\{a_i, a_j^\dagger\} = \delta_{ij}$, $\{a_i, a_j\} = 0$ (item Q1, machine precision).

Theorem 5 (Thirds quantization and the multiplet pattern)**.** *The number operator $N = \sum_k a_k^\dagger a_k$ on $\mathbb{O}\otimes\mathbb{C}$ has spectrum $\{0,1,2,3\}$ with multiplicities $\{1,3,3,1\}$ (item Q2). The $\mathfrak{su}(3)$ clock-unit stabiliser of Paper LVI commutes with N; its restricted image on each 3-dimensional eigenspace spans the full 8-dimensional algebra (item Q3) — a faithfulness certificate: the kernel of a Lie-algebra action is an ideal, $\mathfrak{su}(3)$ is simple (Paper LVI item SM4), and its only nontrivial representations of dimension ≤ 3 are the triplets $\mathbf{3}, \bar{\mathbf{3}}$. The action on each 1-dimensional eigenspace vanishes. Setting*

$$Q \;=\; N/3 \;\in\; \{0,\ \tfrac{1}{3},\ \tfrac{2}{3},\ 1\},$$

*the framework produces a charge-*magnitude *lattice quantized in thirds, in the pattern of one neutral colour singlet, one magnitude-$\frac{1}{3}$ colour triplet, one magnitude-$\frac{2}{3}$ colour triplet, and one magnitude-1 colour singlet — matching $(\nu,\ \bar{d},\ u,\ e^+)$. Charge* signs *and antiparticles live on the conjugate module (complex conjugation exchanges $N \leftrightarrow 3 - N$); promoting the lattice to the full signed electric charge of a dynamical fermion sector additionally requires the doublet assignments and the independent $U(1)$ gauging of §5. Null (item Q4): replacing the J-compatible pairing by an incompatible one destroys the integer spectrum.*

Remark 6 (Provenance and what is added). The construction is the classical octonionic charge algebra of Günaydin–Gürsey [Günaydin and Gürsey, 1973], in the ladder-operator form developed by Furey [Furey, 2016]; see also Baez [2002]. It is re-derived here on the substrate's own Cayley–Dickson table with no representation input, because the programme's claim is not the construction but its *necessity*: the octonionic arena is forced (Adams plus renderability; Papers LIV/LVI), and the distinguished unit is the clock axis the rendering chart already selects. Given those forced inputs, thirds quantization and the $\{1,3,3,1\}$ pattern are not model-building choices — they are what the arena's algebra does.

Corollary 7 (Hypercharge bookkeeping)**.** *For any assignment of the multiplets into weak doublets/singlets, $Y = 2(Q - T_3)$ is determined by Theorem 5's lattice. The bookkeeping half of the hypercharge question is thereby fixed; what remains open is $U(1)_Y$ as an in-*dependent gauge factor *(the embedding/mixing, equivalently the Weinberg angle) — see §5.*

4 What this does to the masses question

This paper closes *kinematic* structure: groups, handedness, charges, multiplets — the exact, algebraic layer. Masses are not kinematic. They are eigenvalues of interacting dynamics, and the framework's current mass content is therefore handled separately and honestly by the parameter-free ledger simulator (`scripts/sm_ledger.py`): anchored only on the muon's structural shell, every particle's integer-shell mass $m = m_\mu \varphi^{96-N}$ is a fixed number, and the per-particle deviations (muon exact by anchor; Z -2.3%; ranging to $\sim 25\%$ at the worst placements) are printed as the framework's current, unfitted error budget. The two derivations that would upgrade PLACED to DERIVED — the shell-integer rule and the offset dynamics — are the named open items; the discipline of not fitting them is what keeps the sharp placements meaningful. The analogy inside established physics is exact: gauge theory fixes charges exactly while leaving masses to dynamics; QED predicts magnetic moments to twelve digits and cannot predict the electron mass at all.

5 The honest list, updated

Closed by this paper, with the scope stated exactly: **chirality as mechanism** (why a left-factor gauge group is one-handed; Corollary 2 with Remark 3's caveats — not the electroweak fermion Lagrangian); **the charge-magnitude lattice and multiplet pattern** (Theorem 5 — not signed dynamical charge, which needs the assignments and gauging below); **hypercharge as arithmetic given assignments** (Corollary 7). Still open, stated plainly:

1. **Independent hypercharge / Weinberg angle.** The derived clock circle lies inside $SU(2)$; the SM's $U(1)_Y$ does not. The charge lattice now constrains any future embedding; the mixing itself has no derivation, and geometric candidate values are scheme-dependent.

2. **Generations.** Named conjecture, untested: the cascade's D_4 rung carries triality (S_3 permuting three 8-dimensional representations), the right *shape* for three families; no mechanism is derived.

3. **Anomaly bookkeeping; doublet assignments.** Arithmetic consistent with standard assignments; not derived.

4. **Masses; shell integers and offsets; neutrino sector; running couplings.** Per §4 and the audit document.

6 Reproducibility

Manuscript: `papers/paper-lviii/paper-lviii.tex`; derivation notes and the closeability audit: `docs/sm-closeability-audit.md`. Verification: `python3 scripts/verify_sm_structure.p` (11/11; NumPy only; the octonions are built in-script by Cayley–Dickson doubling, the quaternionic items use `scripts/600cell_data.npz`). The ledger simulator: `python3 scripts/sm_ledger.py` (no adjustable parameters; prints its own honesty grades). Upstream: Paper LVI (gauge inventory; suite items SM0–SM6), Papers LIII/LVII (frame split, T14; gravity chain).

References

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